

ROCKS

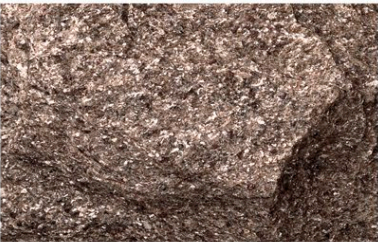
The rocks that form Earth’s crust are composed of different combinations of minerals. Thought of as the epitome of strength and solidity, they are actually in a state of constant change, with rock being destroyed and reformed over immense periods of time. Rocks are classified into three main groups, depending on how they form.



Igneous rocks can be formed by the cooling of magma deep underground, like this granite, or by volcanic eruption.



Sedimentary rocks such as sandstone are formed by the erosion of existing rocks and the deposition of their fragments.



Metamorphic rocks such as schist form when changes in pressure, temperature, or both alter existing rocks.

The world’s oldest known rocks, from Canada’s Northwest Territories, have existed for about 4 billion years, but most rocks are far younger than this. The chalk cliffs facing the English Channel date back to the Cretaceous Period, which ended 66 million years ago (see timeline, below), and the European Alps are younger still. Even in the Grand Canyon, the oldest rocks date back 2 billion years – less than half the lifetime of the planet as a whole. The reason for this is that Earth is tectonically active, with new rock being formed by the planet’s internal heat. At the same time, existing rocks are broken down, in an endless cycle that started when Earth’s crust first solidified.

ROCK GROUPS

Geologists classify rocks into three overall groups – igneous, sedimentary, and metamorphic – reflecting the different ways in which they are formed. Igneous rocks are created by lava from volcanic eruptions on Earth’s surface and from the cooling of molten magma underground. The heat in lava and magma rises from the mantle beneath Earth’s crust. The commonest kind, a black volcanic rock called basalt forms most of the sea floor. Igneous activity can also create plutonic rock, which cools and solidifies beneath the surface, often in huge masses called batholiths. This is how most of the world’s granite originates.

Sedimentary rocks are created on Earth’s surface. Their key feature is their layers, or strata, built up over long periods of time. Some sedimentary rocks – such as sandstone and shale – form when existing rocks are eroded, releasing particles that are then washed or blown away, producing rock elsewhere. Others, such as rock salt and rock gypsum, are created when salt water evaporates, leaving behind its dissolved minerals, known as evaporites. Sedimentary rock can also have a biological origin: chalk is formed from the microscopic skeletons of marine organisms, and coal is derived from the remains of plants, compressed over millions of years. Metamorphism takes place deep beneath Earth’s surface, when rocks are altered by heat, pressure, or both. For example, marble is created when limestone is heated by lava or by magma. Recrystallization of the limestone may remove any bedding in the original rock, enabling some marbles to be cut without splitting apart, a quality prized in sculpture. Metamorphism can take place over very wide areas. If temperature and pressure are sufficiently high, all rocks can be metamorphosed and ultimately melted, forming magma and completing the rock cycle.

GRAND CANYON, USA >

This view of the Grand Canyon shows the almost horizontal layering of sedimentary rocks and the effects of river erosion.

GEOLOGICAL TIMELINE

